

Tianshuo Han

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Education

University of Chinese Academy of Sciences | **Ph.D. Candidate** Sep 2021 — Jan 2027 (Expected)
Institute of Information Engineering | Direct Ph.D. Program

Focused on **Linux kernel** and **operating systems** research, with solid theoretical foundation and extensive hands-on experience in kernel internals, binary analysis, and system-level software testing. Research published at top-tier conferences including IEEE S&P, USENIX, and ACM CCS.

China Agricultural University | **Undergraduate** Sep 2017 — Jun 2021
Biological Sciences, College of Life Sciences | Bachelor of Science

Outstanding academic performance with multiple academic scholarships. Awarded National Second Prize in the China Undergraduate Mathematical Contest in Modeling and Gold Medal in the iGEM Competition.

Publications

NetPanic: A Fuzzer-in-the-Middle Approach to Auditing the Linux Kernel Network Stack

Tianshuo Han, Zong Cao, Zhen Dong, Xiapu Luo, Zhenyu Song, and Jian Liu

47th IEEE S&P, 2026, CCF-A

- Conducted the first systematic fuzzing-based analysis of the Linux kernel network stack. Proposed the Fuzzer-in-the-Middle architecture, achieving over 100x throughput and 400% code coverage improvement over the state-of-the-art, discovering 15 previously unknown kernel bugs.
- Independently completed the overall architecture design and implementation. Led the paper writing effort.

CARDSHARK: Understanding and Stabilizing Linux Kernel Concurrency Bugs Against the Odds

Tianshuo Han, Xiaorui Gong, and Jian Liu

33rd USENIX Symposium, 2024, CCF-A

- Conducted the first in-depth analysis of the root causes of instability in triggering kernel concurrency bugs, and proposed a general-purpose technique that significantly improves the reproducibility and stability of concurrency bug triggering.
- Independently completed the mechanism analysis, technique development, and performance evaluation. Led the paper writing effort.

Reviving Discarded Bugs: Analyzing Previously Unanalyzable Linux Kernel Bugs Through Control Metadata Fields

Hao Zhang, Jie Lu, Shaomin Chen, **Tianshuo Han**, Bolun Zhang, Jian Liu, Xiaorui Gong

32nd ACM CCS, 2025, CCF-A

- Proposed a novel technique based on manipulating critical metadata fields in kernel structures, enhancing analysis capabilities in constrained scenarios and expanding the scope of analyzable kernel bugs.
- Deeply involved in defining the core thesis and responding to reviewer comments. Participated in evaluation experiment design and CodeQL development.

Whole-genome sequence and comparative analysis of *Trichoderma asperellum* ND-1 reveal its unique enzymatic system for efficient biomass degradation

Fengzhen Zheng, **Tianshuo Han**, Abdul Basit, Junquan Liu, Ting Miao, and Wei Jiang

Catalysts 12(4), 2022, IF=4.16

- Revealed the unique lignocellulose degradation enzyme system of *Trichoderma asperellum* ND-1 through whole-genome sequencing and secretome analysis. Responsible for genomic data analysis and visualization.